`lifelines-hc`: Higher Criticism testing for sparse non-proportional hazard departures in Python

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Abstract

The log-rank test is the standard tool for two-sample survival comparison, but it loses power when the hazard difference is confined to a small, unknown subset of the follow-up. Weighted log-rank tests each fix a temporal emphasis in advance; combination procedures such as MaxCombo and the Yang–Prentice short-term/long-term hazard-ratio model scan a small dictionary of global temporal shapes. All remain insensitive to *sparse* departures falling outside those shapes. We present `lifelines-hc`, a Python package that extends the `lifelines` survival library with the HCHG test of Kipnis et al. [2026]: Higher Criticism applied to per-interval hypergeometric p-values. The package exposes a `lifelines`-compatible API, returns interval-level diagnostics identifying which time windows drive a detection, and ships a benchmarking wrapper that runs HCHG alongside the log-rank test, four weighted alternatives, MaxCombo and Yang–Prentice in one call. On four clinical datasets with distinct non-proportional structures, HCHG attains $p \leq 0.014$ throughout where the log-rank test is non-significant everywhere ($p \geq 0.26$), and detects the three departures that MaxCombo and Yang–Prentice miss.

Keywords: survival analysis; higher criticism; sparse signal detection; non-proportional hazards; hypergeometric test; Python.

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Code metadata

Nr.	Code metadata description	Value
C1	Current code version	v0.1.3
C2	Permanent link to code/repository used for this code version	https://github.com/alonkipnis/lifelines-hc/tree/v0.1.3
C3	Permanent link to archived version	https://doi.org/10.5281/zenodo.23170281
C4	Legal code license	MIT
C5	Code versioning system used	git
C6	Software code languages, tools and services used	Python; R (optional, only for the MaxCombo and Yang–Prentice comparators)
C7	Compilation requirements, operating environments, dependencies	Python $\geq$ 3.8; `numpy`, `pandas`, `lifelines` $\geq$ 0.29, `multiple-hypothesis-testing`, `multiHGtest`. Optional: `matplotlib` for plotting; R with `nph` and `YPmodel` for the comparator tests. Platform independent.
C8	Link to developer documentation/manual	https://github.com/alonkipnis/lifelines-hc/blob/v0.1.3/README.md
C9	Support email for questions	alon.kipnis@runi.ac.il

Table 1: Code metadata.

1 Motivation and significance

The Kaplan–Meier estimator [Kaplan and Meier, 1958] and the log-rank test [Mantel, 1966] are the workhorses of two-sample survival analysis. The log-rank statistic has good power against proportional-hazards alternatives [Schoenfeld, 1981], and this is the regime for which it was designed. Kipnis et al. [2026] showed that it is not optimal against *sparse* hazard departures, in which the hazard difference is concentrated in a small, unknown subset of time intervals.

Such departures are common in contemporary trials. Immune checkpoint inhibitors may require an immune-priming phase before producing durable tumour suppression, giving delayed effects and crossing curves: in CheckMate 057, nivolumab improved overall survival relative to docetaxel while the progression-free survival curves crossed [Borghaei et al., 2015]. Targeted therapies can produce benefits confined to a limited window that are diluted in global summaries: in COMET-1, cabozantinib markedly improved bone scan response at week 12 and radiographic progression-free survival without a significant overall survival benefit [Smith et al., 2016]. Subgroup-dependent biology generates composite, time-varying differences: in AZURE, adjuvant zoledronic acid showed no overall benefit but did benefit women with established menopause [Coleman et al., 2011, 2014]. Time-varying treatment effects can produce multi-window departures: in the CSL cirrhosis trial, prednisone was harmful early and beneficial later, yielding curves that cross more than once [Martinussen and Scheike, 2006].

Several weighted log-rank tests address non-proportional hazards [Gehan, 1965, Peto and Peto, 1972, Tarone and Ware, 1977, Harrington and Fleming, 1982, Fleming and Harrington, 1991]. Each applies a temporal weight emphasising early, intermediate or late events, and each has good power when the signal matches its intended pattern — but choosing the weight requires knowing in advance

where the departure will occur. Contemporary practice therefore combines several weights rather than committing to one. MaxCombo [Lin et al., 2020, Ristl et al., 2021] takes the maximum of $\mathrm{FH}(\rho,\gamma)$ statistics over $(\rho,\gamma)\in\{(0,0),(0,1),(1,0),(1,1)\}$ with a multiplicity-adjusted p-value; the Yang–Prentice model [Yang and Prentice, 2005] interpolates between an immediate hazard ratio θ_1 and an asymptotic ratio θ_2, with an associated adaptive weighted log-rank test [Yang and Prentice, 2010]. Both are powerful against the global early-versus-late patterns they were designed for. Both remain averaging methods over a small dictionary of global temporal shapes, and can miss sparse or multi-window departures.

No implementation of a scan-based alternative was available to Python users of `lifelines` [Davidson-Pilon, 2019], the most widely used survival library in that ecosystem. `lifelines-hc` fills that gap: it implements HCHG [Kipnis et al., 2026], requires no a priori temporal weight, and reports which intervals drive a detection — information that no single-number test provides.

2 Software description

2.1 Software architecture

`lifelines-hc` is a pure-Python package of two modules. `lifelines_hc.statistics` implements the interval construction, the exact hypergeometric per-interval p-values, the Higher Criticism aggregation, and the permutation calibration; `lifelines_hc.plotting` provides Kaplan–Meier overlays and p-value profile plots. Every test returns a `lifelines.statistics.StatisticalResult`, the same object returned by `lifelines.statistics.logrank_test`, so results drop into existing `lifelines` workflows and print with the library's own summary formatting. The package adds no state and subclasses nothing; it is a set of functions over NumPy arrays.

Given two groups with right-censored observations (T_i, E_i, G_i), the follow-up range is partitioned into K equal-width intervals. For interval k, let n_k be the total events, r_k^A, r_k^B the subjects at risk in each group, and x_k the events in group B. Under the null of equal group-specific hazards,

$$x_k \mid n_k, r_k^A, r_k^B \;\sim\; \mathrm{Hypergeometric}\big(r_k^A + r_k^B,\; r_k^B,\; n_k\big)\,, \tag{1}$$

yielding an exact one-sided p-value p_k per interval. Ordering these as $p_{(1)} \le \cdots \le p_{(K)}$, the HC statistic [Donoho and Jin, 2004, 2015] is

$$\mathrm{HCHG}_K \;=\; \max_{1\le j\le \lfloor \gamma K+1/2\rfloor} \frac{j/K - p_{(j)}}{\sqrt{(j/K)\,(1-j/K)\,/K}}, \tag{2}$$

where $\gamma\in(0,1)$ (default 0.2) restricts the scan to the most extreme fraction of interval p-values. What is standardised is a deviation of the *empirical* distribution function, not of a single p-value: at $p_{(j)}$ the empirical CDF equals j/K, with null expectation $p_{(j)}$, and K times the empirical CDF at a fixed t is Binomial(K,t), giving the scale $\sqrt{t(1-t)/K}$. Equation (2) evaluates it at $t=j/K$ [Donoho and Jin, 2008]; this is the default, `hc_version="dj2008"`, and is used throughout. The `hc_version` argument also selects `"dj2004"`, which evaluates the same scale at $t=p_{(j)}$ [Donoho and Jin, 2004], and `"beta"`, which standardises the ordered p-value itself by the exact moments of its Beta$(j, K+1-j)$ null distribution. The variants agree asymptotically and differ by $O(K^{-1})$ in finite samples. A global p-value follows from permuting group labels. For the two-sided alternative, the minimum of the two one-sided interval p-values replaces p_k in (2).

Interval pooling is the one parameter requiring user attention. With continuous event times each distinct time carries at most one event, and the hypergeometric p-values are then too coarse to be informative; pooling events into K equal-width intervals via `n_intervals_to_pool` restores

resolution. Values between 50 and 200 work well in practice, and the parameter may be omitted for naturally discrete or heavily tied data.

2.2 Software functionalities

- `higher_criticism_test` — the HCHG test, with exact permutation calibration, one- or two-sided alternatives, optional censoring indicators, and a configurable HC fraction γ.
- `berk_jones_test`, `fisher_combination_test`, `min_p_test` — alternative aggregations of the same per-interval p-values, for sensitivity analysis.
- `event_pvalues` — the raw per-interval p-values.
- `suspected_deviations` — a `pandas.DataFrame` of per-interval p-values with a boolean `suspected` column marking intervals at or below the HC threshold. This is the diagnostic output distinguishing HCHG from a single-number test: it localises the departure.
- `plot_km_with_hc`, `plot_pvalue_profile` — Kaplan–Meier curves with flagged intervals shaded, and signed $-\log_{10}$ p-value profiles.
- `run_all_tests` — a benchmarking wrapper returning a `DataFrame` comparing HCHG, log-rank, Gehan–Wilcoxon, Tarone–Ware, Peto–Prentice, Fleming–Harrington$(1, 1)$, MaxCombo and the Yang–Prentice adaptive log-rank test on one dataset.

```
from lifelines_hc import higher_criticism_test, suspected_deviations

result = higher_criticism_test(
    T_A, T_B,
    event_observed_A=E_A, event_observed_B=E_B,
    n_intervals_to_pool=100, n_permutations=2000,
    alternative='both', gamma=0.2, hc_version='dj2008', seed=42)

print(result.p_value)          # permutation-calibrated p-value
print(result.test_statistic)   # HCHG statistic

flagged = suspected_deviations(T_A, T_B, n_intervals_to_pool=100)
print(flagged[flagged["suspected"]])   # which windows drive the
    signal
```

3 Illustrative examples

We apply `lifelines-hc` to four clinical datasets with distinct non-proportional structures (Figure 1), benchmarking HCHG against the log-rank test, four weighted alternatives, MaxCombo and Yang–Prentice. Throughout, HCHG uses two-sided interval p-values, $\gamma = 0.2$, and 2000 label permutations with the seed fixed at 42. Individual patient data for the first three trials were reconstructed from published Kaplan–Meier curves using the algorithm of Guyot et al. [2012]; the CSL trial provides public individual patient data.

Crossing curves (CheckMate 057). 582 patients with advanced non-squamous NSCLC randomised to nivolumab ($n = 292$) or docetaxel ($n = 290$) [Borghaei et al., 2015]. Progression-free survival curves cross at ∼3 months. The log-rank test is non-significant ($p = 0.351$) as early excess

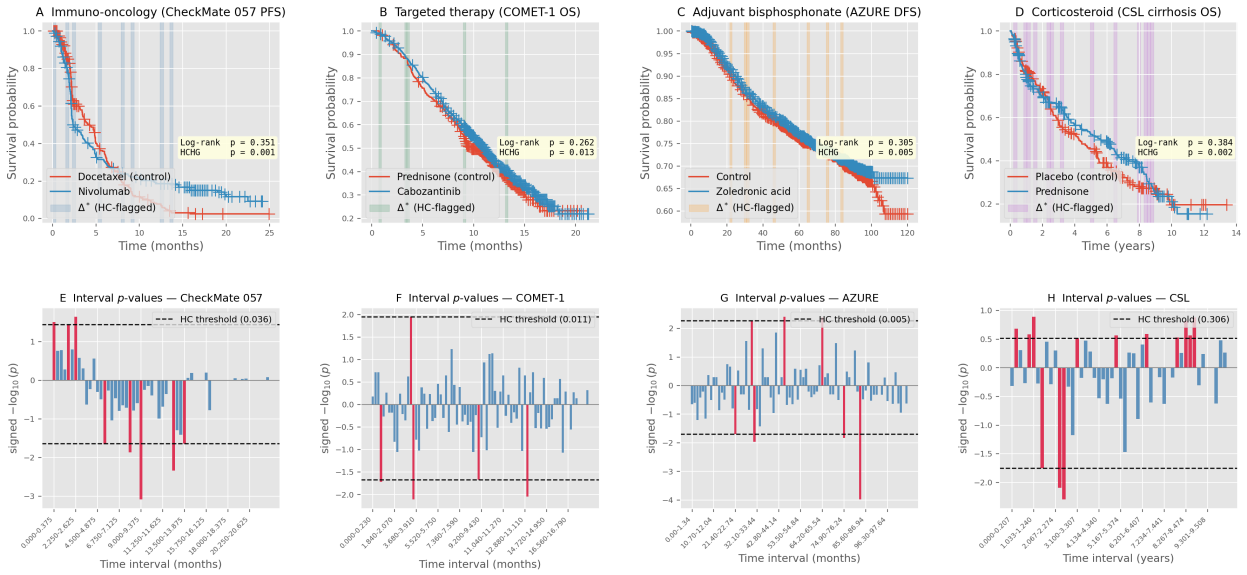


Figure 1: **Higher Criticism detects non-proportional hazard departures across four clinical datasets with distinct temporal patterns.** Top row (A–D): Kaplan–Meier curves with HCHG-flagged intervals shaded. Bottom row (E–H): per-interval *signed* $-\log_{10}$(hypergeometric p-value); upward bars mark excess events in the treatment arm, downward bars excess in the control arm. Dashed lines are the per-direction HCHG thresholds; bars beyond either line are highlighted and coincide with the intervals shaded above. **(A,E)** CheckMate 057 PFS ($n = 582$); log-rank $p = 0.351$, HCHG $p = 0.001$. **(B,F)** COMET-1 OS ($n = 1028$); log-rank $p = 0.262$, HCHG $p = 0.014$. **(C,G)** AZURE DFS ($n = 3359$); log-rank $p = 0.305$, HCHG $p = 0.005$. **(D,H)** CSL OS ($n = 446$, fully public raw data); log-rank $p = 0.384$, HCHG $p = 0.002$.

and late benefit cancel. Gehan–Wilcoxon ($p = 0.041$) and Peto–Prentice ($p = 0.049$) reach borderline significance; Tarone–Ware ($p = 0.358$) and Fleming–Harrington$(1, 1)$ ($p = 0.256$) do not. This is the delayed-benefit pattern MaxCombo and Yang–Prentice target, and both succeed: MaxCombo $p = 2.5 \times 10^{-5}$, driven by the late-weighted FH$(0, 1)$ member ($Z = 4.27$), and Yang–Prentice $p = 0.00055$ with $\hat{\theta}_1 = 2.33$, $\hat{\theta}_2 = 0.44$. HCHG achieves $p = 0.001$ and additionally flags both the early divergence and the late separation windows separately.

Transient early benefit (COMET-1). 1028 patients with chemotherapy-pretreated mCRPC randomised 2:1 to cabozantinib ($n = 682$) or prednisone ($n = 346$) [Smith et al., 2016]. Cabozantinib significantly improved bone scan response at week 12 without a sustained survival benefit, giving a temporally concentrated advantage in months 3–7. The log-rank test gives $p = 0.262$ and all four weighted alternatives are non-significant ($p \geq 0.19$). MaxCombo ($p = 0.338$) and Yang–Prentice ($p = 0.262$) also miss it: a transient few-month advantage is not a global early-versus-late hazard-ratio shape. HCHG gives $p = 0.014$, aggregating evidence across the specific early intervals where the effect is concentrated.

Mid-window accumulation (AZURE). 3359 early-stage breast cancer patients randomised to standard adjuvant therapy with ($n = 1681$) or without ($n = 1678$) zoledronic acid [Coleman et al., 2011, 2014]. The drug's benefit depends on the low-oestrogen postmenopausal state; as premenopausal patients transitioned during the decade-long follow-up, the hazard difference ac-

cumulated in months 20–60 rather than uniformly. Log-rank $p = 0.305$, all weighted alternatives $p \geq 0.28$, MaxCombo $p = 0.401$, Yang–Prentice $p = 0.308$. HCHG gives $p = 0.005$. The detection is corroborated by the published subgroup analyses showing significant benefit in postmenopausal patients, indicating a genuine biological effect rather than an artefact.

Multi-window, sign-changing effect (CSL). 446 patients with histologically verified liver cirrhosis randomised to prednisone ($n = 226$) or placebo ($n = 220$), with 270 deaths over up to ∼13 years [Martinussen and Scheike, 2006]. Unlike the preceding three, the raw individual patient data are public, via the CRAN `timereg` package and the SurvSet repository [Drysdale, 2022]; this example is therefore reproducible end to end from public sources. Corticosteroids cause early harm followed by later anti-inflammatory benefit, and the curves cross more than once. Log-rank $p = 0.384$; Gehan–Wilcoxon $p = 0.51$, Tarone–Ware $p = 0.38$, Peto–Prentice $p = 0.45$, Fleming–Harrington$(1, 1)$ $p = 0.14$; MaxCombo $p = 0.236$, Yang–Prentice $p = 0.393$. HCHG gives $p = 0.002$: no single temporal weight, and no two-parameter short-term/long-term hazard ratio, represents a multi-window sign-changing departure.

Summary. Across the four datasets HCHG attains $p \leq 0.014$ in every case while the log-rank test is non-significant throughout ($p \geq 0.26$). MaxCombo and Yang–Prentice are significant only on CheckMate 057, the delayed-benefit crossing they were designed to detect; on the other three they return $p \geq 0.24$. HCHG is therefore complementary rather than merely competitive: it matches the combination procedures on global early-versus-late non-proportionality and retains power where a small Fleming–Harrington dictionary cannot reach.

4 Impact

`lifelines-hc` makes a published but previously unimplemented test available to the Python survival-analysis community. Before this package, applying HCHG required reimplementing the interval construction, the exact hypergeometric calculation and the permutation calibration from the method paper — enough friction that the test was effectively unavailable outside the authors' own code.

Three consequences follow for practice. First, trial analysts working in Python gain a non-proportional-hazards test that requires no a priori commitment to a temporal weight, complementing rather than replacing the MaxCombo and Yang–Prentice procedures that regulatory-facing work currently relies on. Second, the `suspected_deviations` output localises a detection to specific follow-up windows, which turns a significance result into a hypothesis about mechanism: in the AZURE example the flagged window corresponds to the menopausal transition, and in CSL to the switch from early harm to late benefit. Third, `run_all_tests` gives a single call that benchmarks eight procedures on one dataset, lowering the cost of reporting a non-proportional-hazards analysis against the full comparator set rather than against the log-rank test alone.

Because every function returns a `lifelines StatisticalResult`, adoption requires no change to an existing analysis pipeline beyond the import. The package is on PyPI, is MIT licensed, and carries a test suite of 24 cases covering interval construction and pooling, censoring handling, one- and two-sided alternatives, permutation calibration and seed reproducibility.

5 Conclusions

`lifelines-hc` provides an accessible Python implementation of HCHG for two-sample right-censored survival data. HCHG targets sparse hazard departures at unknown temporal locations: unlike weighted log-rank alternatives, MaxCombo and the Yang–Prentice adaptive test, it requires no a priori specification of where the hazard difference will occur, and it returns interval-level information identifying which windows drive the signal. Across four clinical datasets with distinct non-proportional structures it detects departures that the log-rank test misses entirely and that the combination procedures miss in three cases out of four.

Planned work includes an R port for distribution via CRAN, where the survival-analysis community outside Python primarily works, and extension of the interval construction to adaptive rather than equal-width binning.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Individual patient data for CheckMate 057, COMET-1 and AZURE were reconstructed from published Kaplan–Meier curves using the algorithm of Guyot et al. [2012] as implemented in the `kmdata` R package (`https://github.com/raredd/kmdata`). CSL liver-cirrhosis individual patient data are public via the CRAN `timereg` package (dataset `csl`) and the SurvSet repository [Drysdale, 2022]. All analysis code is available at `https://github.com/alonkipnis/lifelines-hc/tree/v0.1.3/AppNote`.

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